

HDOC Day-19

Implementation of SGPA and CGPA Calculation Using Nested Loops & Decision-Making Constructs in C

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Aim

To write and test a C program that calculates the SGPA of each semester and the final CGPA for multiple semesters using nested loops and decision-making statements.

Grade Conversion Scale

Grade	O	A	B	C	D	E	P
Grade Point	10	9	8	7	6	5	4

Algorithm (Simplified from the Given Algorithm)

1. Start.
2. Read global ID and keep asking until a positive value is entered.
3. Read the number of semesters (m) and validate that it is positive.
4. Set cumulative total credits and cumulative total grade credits to 0.
5. For each semester, set semester total credits and semester grade credits to 0.
6. Read the number of courses (n) and validate it.
7. For every course, read and validate course code, credit and grade.
8. Convert the grade to grade point: O=10, A=9, B=8, C=7, D=6, E=5, P=4.
9. Add (grade point x credit) to semester grade credits and add credit to semester credits.
10. SGPA = semester grade credits / semester credits. Display the semester SGPA.
11. Add semester totals to cumulative totals.
12. CGPA = cumulative grade credits / cumulative credits. Display Global ID, total credits and CGPA.
13. Stop.

Test Case Table

No.	Input / Condition	Type	Expected Result
1	ID=590032925, m=1, n=2; (101,4,A), (102,3,B)	Valid	SGPA is calculated correctly
2	Global ID = 0, then 590032925	Invalid	Error shown; asks for ID again
3	m = -2, then 1	Invalid	Error shown; asks for semesters again
4	n = 0, then 1	Boundary / Invalid	Error shown; asks for courses again
5	Course code = -10, then 101	Invalid	Error shown; asks for code again
6	Credit = 0, then 4	Boundary / Invalid	Error shown; asks for credit again

No.	Input / Condition	Type	Expected Result
7	Grade = X, then O	Invalid	Error shown; asks for grade again
8	m=1, n=1, credit=1, grade=P	Boundary valid	SGPA = 4.00
9	m=1, n=1, credit=1, grade=O	Boundary valid	SGPA = 10.00

C Program

```
#include <stdio.h>

int main() {
    int global_id, m, n, semester, course;
    int code, credit, GP;
    int semester_TC, semester_TGC;
    int cumulative_TC = 0, cumulative_TGC = 0;
    char grade;
    float SGPA, CGPA;

    do {
        printf("Enter Global ID: ");
        scanf("%d", &global_id);
        if (global_id <= 0)
            printf("Invalid Global ID. Enter again.\n");
    } while (global_id <= 0);

    do {
        printf("Enter number of semesters: ");
        scanf("%d", &m);
        if (m <= 0)
            printf("Invalid number of semesters. Enter again.\n");
    } while (m <= 0);

    for (semester = 1; semester <= m; semester++) {
        semester_TC = 0;
        semester_TGC = 0;

        printf("\nSemester %d\n", semester);

        do {
            printf("Enter number of courses: ");
            scanf("%d", &n);
            if (n <= 0)
                printf("Invalid number of courses. Enter again.\n");
        } while (n <= 0);

        for (course = 1; course <= n; course++) {
            printf("\nCourse %d\n", course);

            do {
                printf("Enter course code: ");
                scanf("%d", &code);
                if (code <= 0)
                    printf("Invalid course code. Enter again.\n");
            } while (code <= 0);

            do {
                printf("Enter credit: ");
                scanf("%d", &credit);
                if (credit <= 0)
                    printf("Invalid credit. Enter again.\n");
            } while (credit <= 0);

            do {
                printf("Enter grade (O/A/B/C/D/E/P): ");
                scanf(" %c", &grade);

                switch (grade) {
                    case 'O': GP = 10; break;
                    case 'A': GP = 9; break;
                    case 'B': GP = 8; break;
                    case 'C': GP = 7; break;
                    case 'D': GP = 6; break;
                    case 'E': GP = 5; break;
                    case 'P': GP = 4; break;
                    default:
                        GP = -1;
                        printf("Invalid grade. Enter again.\n");
                }
            } while (GP == -1);

            semester_TGC = semester_TGC + (GP * credit);
            semester_TC = semester_TC + credit;
        }

        SGPA = (float)semester_TGC / semester_TC;
        printf("\nSemester %d SGPA = %.2f\n", semester, SGPA);
        cumulative_TGC = cumulative_TGC + semester_TGC;
    }
}
```

```
        cumulative_TC = cumulative_TC + semester_TC;
    }

    CGPA = (float)cumulative_TGC / cumulative_TC;

    printf("\n----- FINAL RESULT -----\\n");
    printf("Global ID      : %d\\n", global_id);
    printf("Semesters       : %d\\n", m);
    printf("Total Credits  : %d\\n", cumulative_TC);
    printf("CGPA          : %.2f\\n", CGPA);

    return 0;
}
```

Sample Execution

```
Enter Global ID: 590032925
Enter number of semesters: 2

Semester 1
Enter number of courses: 2

Course 1
Enter course code: 101
Enter credit: 4
Enter grade (0/A/B/C/D/E/P): A

Course 2
Enter course code: 102
Enter credit: 3
Enter grade (0/A/B/C/D/E/P): B

Semester 1 SGPA = 8.57

Semester 2
Enter number of courses: 2

Course 1
Enter course code: 201
Enter credit: 4
Enter grade (0/A/B/C/D/E/P): 0

Course 2
Enter course code: 202
Enter credit: 3
Enter grade (0/A/B/C/D/E/P): A

Semester 2 SGPA = 9.57

----- FINAL RESULT -----
Global ID      : 590032925
Semesters      : 2
Total Credits  : 14
CGPA           : 9.07
```

Sample Calculation

Semester 1: $(4 \times 9) + (3 \times 8) = 60$ grade credits; total credits = 7; $SGPA = 60/7 = 8.57$.

Semester 2: $(4 \times 10) + (3 \times 9) = 67$ grade credits; total credits = 7; $SGPA = 67/7 = 9.57$.

CGPA: $(60 + 67) / (7 + 7) = 127/14 = 9.07$.

Result

The C program was prepared using nested loops and decision-making constructs. It validates the required inputs, calculates the SGPA for each semester, and calculates the final CGPA using cumulative credits.